Example 3.25

Solve the equation $x^3 - 5x^2 - 4x + 20 = 0$.

Solution

If $P(x)$ denotes the polynomial in the equation, then $P(2) = 0$. Hence 2 is a root of the polynomial equations. To find other roots, we divide the given polynomial $x^3 - 5x^2 - 4x + 20$ by $x - 2$ and get $Q(x) = x^2 - 3x - 10$ as the quotient. Solving $Q(x) = 0$ we get -2 and 5 as roots. Thus $2, -2, 5$ are the solutions of the given equation. ■

Guessing a number as a root by trial and error method is not an easy task. But when the coefficients are integers, using its leading coefficient and the constant term, we can list certain rational numbers as possible roots. Rational Root Theorem helps us to create such a list of possible rational roots. We recall that if a polynomial has rational coefficients, then by multiplying by suitable numbers we can obtain a polynomial with integer coefficients having the same roots. So we can use **Rational Root Theorem**, given below, to guess a few roots of polynomial with rational coefficient. We state the theorem without proof.

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Theorem 3.5 (Rational Root Theorem)

Let $a_n x^n + \cdots + a_1 x + a_0$ with $a_n \neq 0$ and $a_0 \neq 0$, be a polynomial with integer coefficients. If $\frac{p}{q}$, with $(p, q) = 1$, is a root of the polynomial, then p is a factor of a_0 and q is a factor of a_n.

When $a_n = 1$, if there is a rational root $\frac{p}{q}$, then as per theorem 3.5 q is a factor of a_n, then we must have $q = \pm 1$. Thus p must be an integer. So a monic polynomial with integer coefficient cannot have non-integral rational roots. So when $a_n = 1$, if at all there is a rational root, it must be an integer and the integer should divide a_0. (We say an integer a divides an integer b, if $b = ad$ for some integer d.)

As an example let us consider the equation $x^2 - 5x - 6 = 0$. The divisors of 6 are $\pm 1, \pm 2, \pm 3, \pm 6$. From Rational Root Theorem, we can conclude that $\pm 1, \pm 2, \pm 3, \pm 6$ are the only possible solutions of the equation. It does not mean that all of them are solutions. The two values -1 and 6 satisfy the equation and other values do not satisfy the equation.

Moreover, if we consider the equation $x^2 + 4 = 0$, according to the Rational Root theorem, the possible solutions are $\pm 1, \pm 2, \pm 4$; but none of them is a solution. The Rational Root Theorem helps us only to guess a solution and it does not give a solution.

Example 3.26

Find the roots of $2x^3 + 3x^2 + 2x + 3 = 0$.

Solution

According to our notations, $a_n = 2$ and $a_0 = 3$. If $\frac{p}{q}$ is a zero of the polynomial, then as $(p, q) = 1$, p must divide 3 and q must divide 2. Clearly, the possible values of p are $1, -1, 3, -3$ and the possible values of q are $1, -1, 2, -2$. Using these p and q we can form only the fractions

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